4 Harmonic Expansion and Time Rigidity

4.1 Generator-Relative Setup

Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a finite symmetric set of generators ($\Gamma = \Gamma^{-1}$, $|\Gamma| = d$), and let $G = \langle \Gamma \rangle \leq \mathrm{SL}_2(\mathbb{Z})$ be the group they generate.

For a prime $p > 3$, let $\rho_p \colon \mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{F}_p)$ be the reduction-mod-p homomorphism, and define the Cayley graph

$$\mathcal{G}_p \;=\; \mathrm{Cay}\big(\rho_p(G),\ \rho_p(\Gamma)\big).$$

Definition 4.1 (Reachability time and spectral gap)**.** *The* reachability time *from x to y in $\mathcal{G}_p$ is*

$$T_\Gamma(x \to y) \;=\; \min\big\{k : y = g_k \cdots g_1 x,\ g_i \in \rho_p(\Gamma)\big\}.$$

The diameter *is* $\mathrm{diam}(\mathcal{G}_p) = \max_{x,y} T_\Gamma(x \to y)$*. The* spectral gap γ_p *is the second-smallest eigenvalue of the normalized Laplacian $L_p = I - \frac{1}{d} A_p$, where A_p is the adjacency matrix of $\mathcal{G}_p$ (valid since $\mathcal{G}_p$ is d-regular when Γ acts freely).*

Remark (Prime lift and CRT splitting)**.** *For applications to the mod-24 harmonic seed families, we set $M_p = 24p$ with $p > 3$ prime, so* $\gcd(24, p) = 1$*. By the Chinese Remainder Theorem,*

$$(\mathbb{Z}/M_p\mathbb{Z})^2 \;\cong\; (\mathbb{Z}/24\mathbb{Z})^2 \times (\mathbb{Z}/p\mathbb{Z})^2.$$

The factor $(\mathbb{Z}/24\mathbb{Z})^2$ carries the mod-24 resonance structure (fixed attractor families, parity layers) and is independent of p. The factor $(\mathbb{Z}/p\mathbb{Z})^2 \cong \mathbb{F}_p^2$ carries the large-scale geometric structure studied in this section. All spectral and diameter statements below refer to the $\mathbb{F}_p^2$ factor.

4.2 Harmonic Structure Definitions

Definition 4.2 (Harmonic foliation)**.** *A* harmonic foliation *of the finite set $V = \mathbb{F}_p^2$ is a nontrivial Γ-invariant partition $V = \bigsqcup_i F_i$ (nontrivial means $|V| > |\{F_i\}| > 1$).*

Examples: fixed-parity sublattices; fixed coordinate lines; diagonal cosets; fixed mod-q residue classes for $q \mid p - 1$.

Definition 4.3 (Harmonic reducibility)**.** *The pair (V, Γ) is* harmonically reducible *if a harmonic foliation exists; otherwise it is* harmonically irreducible.

Remark. *Harmonic reducibility of* (V, Γ) *is equivalent to the linear group* $\rho_p(G) \leq \mathrm{GL}_2(\mathbb{F}_p)$ *preserving a nontrivial subspace of* $\mathbb{F}_p^2$*, i.e., the action of* $\rho_p(G)$ *on* $\mathbb{F}_p^2$ *is* reducible *in the representation-theoretic sense.*

4.3 Corrected Core Structure Lemma

The following replaces the erroneous non-amenability claim from earlier drafts. Every finite group is amenable; the correct dichotomy lives at the level of the *infinite* group $G \leq \mathrm{SL}_2(\mathbb{Z})$.

Lemma 4.4 (Tits Alternative, [4])**.** *Let* $G \leq \mathrm{SL}_2(\mathbb{Z})$ *be finitely generated. Then exactly one of the following holds:*

(A) *G is virtually solvable.*

(B) *G contains a free subgroup F_2 on two generators.*

Proof. $\mathrm{SL}_2(\mathbb{Z}) \leq \mathrm{GL}_2(\mathbb{Q})$, and $\mathbb{Q}$ has characteristic zero, so the Tits Alternative for finitely generated linear groups over characteristic-zero fields [4] applies directly. □

Remark (Spectral gap, not amenability)**.** *Every finite group is amenable; each* $\mathrm{SL}_2(\mathbb{F}_p)$ *is therefore trivially amenable. Amenability carries no information distinguishing expansion behavior among the Cayley graphs* $\{\mathcal{G}_p\}$*. The correct invariant is the* uniform spectral gap *of the family:* $\inf_p \gamma_p > 0$ *(expansion) versus* $\gamma_p \to 0$ *(non-expansion). This is a property of the* family $\{\mathcal{G}_p\}$*, controlled by the structure of the infinite lifting group* $G \leq \mathrm{SL}_2(\mathbb{Z})$*, not by amenability of individual finite quotients.*

4.4 Harmonic Expansion Dichotomy

Definition 4.5 (Zariski-dense subgroup)**.** $G \leq \mathrm{SL}_2(\mathbb{Z})$ *is* Zariski-dense *in* SL_2 *if its Zariski closure in* $\mathrm{SL}_2(\mathbb{C})$ *(in the Zariski topology of* $\mathbb{A}^4$*) equals* $\mathrm{SL}_2(\mathbb{C})$*. Equivalently, G is not contained in any proper algebraic subgroup of* SL_2 *over* $\overline{\mathbb{Q}}$*.*

Lemma 4.6 (Zariski density $\Leftrightarrow$ case (B))**.** $G \leq \mathrm{SL}_2(\mathbb{Z})$ *is Zariski-dense if and only if G is not virtually solvable (equivalently, case* (B) *of Lemma 4.4 holds).*

Proof. Virtually solvable subgroups of SL_2 are contained in a Borel subgroup (triangularizable over $\overline{\mathbb{Q}}$), hence have Zariski closure $\subsetneq \mathrm{SL}_2$. Conversely, if G is not virtually solvable it contains F_2; by the Tits proof, $F_2 \leq \mathrm{SL}_2(\mathbb{Z})$ is Zariski-dense. □

Theorem 4.7 (Harmonic Expansion Dichotomy)**.** *Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a finite symmetric generating set, $G = \langle \Gamma \rangle$. For each prime $p > 3$, let $\mathcal{G}_p = \mathrm{Cay}(\rho_p(G), \rho_p(\Gamma))$ with spectral gap γ_p. Then:*

(A) ***Virtually solvable case.*** *If G is virtually solvable, then G has polynomial growth, and the family $\{\mathcal{G}_p\}$ is* not *an expander: $\gamma_p \to 0$ as $p \to \infty$.*

(B) ***Zariski-dense / free-core case.*** *If G is Zariski-dense in SL_2, then:*

  (i) *G contains a free subgroup F_2 (Lemma 4.4).*

  (ii) *$\rho_p(G) = \mathrm{SL}_2(\mathbb{F}_p)$ for all but finitely many primes p (strong approximation, [3]).*

  (iii) *There exists $\varepsilon > 0$ such that $\gamma_p \geq \varepsilon$ for all sufficiently large p: the family $\{\mathcal{G}_p\}$ is an* expander family *([1]).*

  (iv) $\mathrm{diam}(\mathcal{G}_p) = O(\log p)$.

Proof. Case (A). Virtual solvability implies polynomial growth (Milnor–Wolf theorem). Polynomial-growth groups satisfy Kesten's criterion for amenability, from which $\gamma_p \to 0$ as the quotient grows.

Case (B)-(i). Lemma 4.4. The free subgroup $F_2 \leq G$ alone does *not* imply expansion of the finite quotients. Expander families are a property of finite Cayley-graph sequences; free subgroup existence in the lifting group G is necessary but not sufficient. Uniform spectral gap of $\{\mathcal{G}_p\}$ requires the arithmetic input provided by items (ii) and (iii): surjectivity of ρ_p (strong approximation) and the product-growth machinery of Bourgain–Gamburd [1].

Case (B)-(ii). Since G is Zariski-dense, the strong-approximation theorem of Matthews–Vaserstein–Weisfeiler [3] implies $\rho_p(G) = \mathrm{SL}_2(\mathbb{F}_p)$ for all but finitely many p.

Case (B)-(iii). We invoke Bourgain–Gamburd [1] under the following explicit hypotheses, all of which hold in our setting:

(H1) $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ is a *fixed* finite symmetric generating set (independent of p).

(H2) $G = \langle \Gamma \rangle$ is *Zariski-dense* in SL_2 over $\overline{\mathbb{Q}}$ (Definition 4.5).

(H3) For all sufficiently large primes p, $\rho_p(G) = \mathrm{SL}_2(\mathbb{F}_p)$ (established in item (ii) via strong approximation).

Under (H1)–(H3), Bourgain–Gamburd guarantees the existence of $\varepsilon > 0$ such that $\gamma_p \geq \varepsilon$ for all sufficiently large p: the family $\{\mathcal{G}_p\}$ is an expander family.

Case (B)-(iv). From (iii), the spectral gap $\gamma_p \geq \varepsilon > 0$. A d-regular expander on n vertices has diameter $O(\log_d n)$ (standard; see e.g. [2]). Here $n = |\mathrm{SL}_2(\mathbb{F}_p)| = p(p^2-1) = O(p^3)$, giving $\mathrm{diam}(\mathcal{G}_p) = O(\log p)$. □

Remark (No entropy; correct observables)**.** *Earlier drafts claimed $h_M \sim 1/M$ as a "harmonic entropy." This is incorrect: the topological entropy of any finite deterministic system is zero. The correct observable characterizing expansion is the spectral gap γ_p (or equivalently the Cheeger constant via the discrete Cheeger–Buser inequality), and the correct scaling observable is* $\mathrm{diam}(\mathcal{G}_p)$*. These replace all entropy claims throughout.*

4.5 No Continuous-Time Obstruction — Corrected Statement

Definition 4.8 (Polynomial-volume scaling limit)**.** *A family $\{(X_p, d_p)\}_{p\to\infty}$ of metric spaces has a* polynomial-volume scaling limit *if there exist scale factors $s_p > 0$ such that $(X_p, d_p/s_p)$ converges, in the pointed Gromov–Hausdorff sense, to a metric measure space (Y, d_Y, μ) whose volume satisfies*

$$\mu\big(B(y,r)\big) \;\leq\; Cr^n$$

for some constants $C, n > 0$, all $y \in Y$, and all $r > 0$.

This class includes: connected Riemannian manifolds (Bishop–Gromov), flat tori, and Carnot groups (Pansu's volume formula). It is the natural target for rescaling limits of polynomial-growth groups. Fractal and p-adic spaces are excluded.

For our setting: X_p is the vertex set of $\mathcal{G}_p$, $d_p = T_\Gamma$ is the graph distance (reachability time), and we ask whether any sequence s_p produces such a limit.

Lemma 4.9 (Growth obstruction)**.** *Let $\{(X_p, d_p)\}_{p\geq p_0}$ be a family of connected d-regular graphs with uniform spectral gap $\gamma_p \geq \varepsilon > 0$. Then:*

(i) ***Expander ball growth.*** *There exists $c = c(\varepsilon, d) > 0$ such that*

$$|B(x,k)| \;\geq\; (1+c)^k$$

for all vertices $x \in X_p$ and all $1 \leq k \leq \frac{1}{2}\mathrm{diam}(X_p)$.

(ii) ***No polynomial-volume GH limit.*** *For every sequence $s_p > 0$, the rescaled family $(X_p,\, d_p/s_p)$ does* not *Gromov–Hausdorff converge to any metric measure space with polynomial volume growth.*

Proof. (i). Uniform spectral gap $\gamma_p \geq \varepsilon$ implies, via the discrete Cheeger–Buser inequality, that the edge-expansion constant $h(X_p) \geq \varepsilon/2$. Standard ball-growth estimates for graphs with positive Cheeger constant give exponential growth with $c \geq \varepsilon/(2d)$.

(ii). Suppose $(X_p, d_p/s_p) \xrightarrow{GH} (Y, d_Y, \mu)$ where $\mu(B(y,r)) \leq Cr^n$ for some $C, n > 0$. For GH convergence to be nondegenerate we need $s_p = \Theta(\mathrm{diam}(X_p))$. For any $r > 0$, the vertex count within rescaled distance r satisfies $|B(x, rs_p)| \geq (1+c)^{rs_p}$, which grows without bound in p, contradicting polynomial growth of μ. □

Theorem 4.10 (Generator Growth Dichotomy)**.** *Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be finite symmetric, $G = \langle \Gamma \rangle$. Exactly one of the following holds:*

(A) ***Polynomial growth regime.*** *G is virtually solvable. Ball volumes in $\mathcal{G}_p$ grow polynomially in radius, and the family admits a polynomial-volume scaling limit.*

(B) ***Exponential growth regime.*** *G is Zariski-dense. There exists $c = c(\Gamma) > 0$ such that for all sufficiently large primes p,*

$$\left|B_{\mathcal{G}_p}(x,r)\right| \;\geq\; e^{cr}$$

for all vertices x and all $1 \leq r \leq C \log p$. The family admits no *polynomial-volume scaling limit.*

The growth regime is determined solely by the Zariski closure of G in SL_2.

Proof. Case (A). Virtual solvability in $\mathrm{SL}_2(\mathbb{Z})$ implies virtual abelianness: the proper algebraic subgroups of SL_2 are Borel (unipotent) or split torus, both virtually abelian. Virtually abelian groups have polynomial growth (Milnor–Wolf theorem). The congruence quotients $G_p = \rho_p(G)$ are abelian of bounded rank; their Cayley graphs have polynomial ball growth (radius-r ball has size $O(r^{\mathrm{rank}})$), and rescale by $s_p = |G_p|^{1/\mathrm{rank}}$ to flat Riemannian tori, which have polynomial volume growth. A polynomial-volume scaling limit exists.

Case (B). G Zariski-dense $\Rightarrow$ by Theorem 4.7(B)(iii), $\gamma_p \geq \varepsilon > 0$ for all large p. The discrete Cheeger–Buser inequality gives $h(\mathcal{G}_p) \geq \varepsilon/2$, whence $|B(x,r)| \geq (1+c')^r$ with $c' = \varepsilon/(2d)$; set $c = \log(1+c') > 0$. This is the exponential lower bound.

For the no-limit claim: suppose (Y, d_Y, μ) with $\mu(B(y,r)) \leq Cr^n$ is a GH limit of the rescaled family. By Theorem 4.7(B)(iv), $s_p = \Theta(\log p)$. For any fixed $r > 0$,

$$|B(x, rs_p)| \;\geq\; e^{crs_p} \;\to\; \infty \quad (p \to \infty),$$

contradicting polynomial volume growth of μ. $\square$

Corollary 4.11 (No Continuous-Time Obstruction)**.** *In the exponential growth regime (Case* (B) *of Theorem 4.10), the family* $\{(\mathcal{G}_p, T_\Gamma)\}$ *admits no polynomial-volume scaling limit. In particular, it admits no Riemannian and no sub-Riemannian (Carnot) scaling limit.*

Remark (Role of spectral gap)**.** *The spectral gap* $\gamma_p \geq \varepsilon > 0$ *(Bourgain–Gamburd) is used in Case (B) as an intermediate step: Zariski-dense* $\Rightarrow$ *uniform gap* $\Rightarrow$ *exponential ball growth. It is not the structural invariant of the theorem. The invariant is the* growth class *of the generator group, which cleanly separates polynomial-volume limits from non-existence of any such limit, across both Riemannian and sub-Riemannian target spaces.*

4.6 NCTO as a Generator-Relative Obstruction Principle

Theorem 4.10 characterises growth class from Γ alone. The implication chain below extracts the spectral and geometric consequences in computable form.

Definition 4.12 (Generator obstruction tag)**.** *A finite symmetric* $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ *carries the* NCTO tag

$$\tau(\Gamma) \;=\; \begin{cases} \texttt{ncto-blocked} & \text{if } \langle\Gamma\rangle \text{ is Zariski-dense in } \mathrm{SL}_2, \\ \texttt{ok} & \text{if } \langle\Gamma\rangle \text{ is virtually solvable.} \end{cases}$$

By Lemma 4.4, exactly one case applies to any finitely generated $G \leq \mathrm{SL}_2(\mathbb{Z})$.

Theorem 4.13 (NCTO, Generator-Relative Implications)**.** *Let* $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ *be finite symmetric,* $G = \langle\Gamma\rangle$*. Then:*

$$\textit{(i)} \;\Rightarrow\; \textit{(ii)} \;\Rightarrow\; \textit{(iii)} \;\Rightarrow\; \textit{(iv)},$$

where:

(i) $\tau(\Gamma) = \texttt{ncto-blocked}$ *(G Zariski-dense).*

(ii) *There exists* $\varepsilon = \varepsilon(\Gamma) > 0$ *such that* $\gamma_p \geq \varepsilon$ *for all sufficiently large* p.

(iii) $\mathrm{diam}(\mathcal{G}_p) = O(\log p)$.

(iv) $\{(\mathcal{G}_p, T_\Gamma)\}$ *has no polynomial-volume scaling limit (Definition 4.8).*

Moreover, $\varepsilon(\Gamma)$ *depends only on the Zariski closure of* G*; for* $\Gamma = \{\sigma^{\pm 1}, w^{\pm 1}\}$ *it is made explicit by Bourgain–Gamburd [1].*

Proof. (i) ⇒ (ii). Theorem 4.7(B)(iii): Bourgain–Gamburd under (H1)–(H3).

(ii) ⇒ (iii). A d-regular expander on n vertices has diameter $O(\log n)$ (see e.g. [2]). Here $n = |\mathrm{SL}_2(\mathbb{F}_p)| = O(p^3)$, giving $O(\log p)$.

(iii) ⇒ (iv). Logarithmic diameter implies exponential ball growth (Lemma 4.9(i)), which prevents GH convergence to any polynomial-volume space (Lemma 4.9(ii)). □

Remark (Converses fail)**.** *The implications (ii)⇒(iii) and (iii)⇒(iv) are strict. (ii) ⇏ (iii): there exist bounded-degree graph families with diameter $O(\log n)$ and spectral gap $\gamma_p \to 0$; logarithmic diameter alone does not force a spectral gap lower bound. (i) ⇏ (iv): Definition 4.8 already includes Carnot (sub-Riemannian) limits, so virtually nilpotent groups (whose Pansu limits are Carnot groups) are correctly placed on the "limit exists" side. However, (iv) alone does not force Zariski-density without the growth classification of Theorem 4.10: one must also rule out other reasons for non-convergence (e.g. non-compactness of the limit).*

Remark (QA obstruction types)**.** *In the QA generator-certificate framework, the NCTO tag classifies generators by the* type *of geometric obstruction they carry:*

- ***Type I (topological):*** *connectivity or reachability obstruction; the Cayley graph is disconnected or has isolated vertices.*
- ***Type II (metric / PDI):*** *route-reducing obstruction; the generator is tagged* `parity-block`*, reducing multi-path states and lowering the Path Diversity Index.*
- ***Type III (spectral / NCTO):*** *expansion obstruction. Tag: $\tau(\Gamma) =$* `ncto-blocked`*. Effect: uniform gap $\varepsilon(\Gamma) > 0$, no GH Riemannian limit.*

Unlike Type I and Type II obstructions, which are local (they depend on individual generator edges), the Type III obstruction is global and family-level*: it is a property of the entire congruence quotient family $\{\mathcal{G}_p\}$, determined by the Zariski closure of $G = \langle\Gamma\rangle$ in SL_2.*

The tag $\tau(\Gamma)$ is finitely decidable *from Γ: algorithms for testing virtual solvability in $\mathrm{SL}_2(\mathbb{Z})$ are known. In practice, reduction modulo several primes provides strong numerical evidence (if $\rho_p(\langle\Gamma\rangle) = \mathrm{SL}_2(\mathbb{F}_p)$ for multiple primes, Zariski-density is highly indicated), but a single prime test does not constitute a proof of the Bourgain–Gamburd hypotheses across all large p.*

4.7 Generator Specification and Correction

Definition 4.14 (Prime-lifted tonal action)**.** *Let $N = 24$ be the chromatic period of the mod-24 QA harmonic seed system, and let $p > 3$ be prime with $M_p = 24p$. The* prime-lifted tonal action *is the action of $G_0 = \langle \sigma, w \rangle = \mathrm{SL}_2(\mathbb{Z})$ on $(\mathbb{Z}/M_p\mathbb{Z})^2$ via entry-wise reduction mod M_p.*

The two basic tonal operations are identified as follows:

- ***Semitone transposition**: $\sigma = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ shifts the "base" coordinate by one step mod M_p, corresponding to ascending one semitone in the 24-tone chromatic lattice.*
- ***Melodic inversion**: $w = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$ (Weyl element) swaps and negates coordinates, corresponding to inversion of a pitch-class set around the central axis.*

By the CRT splitting $(\mathbb{Z}/M_p\mathbb{Z})^2 \cong (\mathbb{Z}/24\mathbb{Z})^2 \times \mathbb{F}_p^2$ (Remark 4.1), the mod-24 factor records the harmonic seed structure (attractor families, parity layers), while the $\mathbb{F}_p^2$ factor is the Cayley-graph domain analyzed by Bourgain–Gamburd. As $p \to \infty$, the lattice resolution increases, and the spectral geometry of $\mathcal{G}_p$ (Theorem 4.7) governs the large-scale tonal structure.

Earlier drafts used generators σ, μ, λ where $\mu = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ has $\det \mu = -1 \notin \mathrm{SL}_2(\mathbb{Z})$. The corrected generator set, all lying in $\mathrm{SL}_2(\mathbb{Z})$, is:

$$\sigma = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \lambda_k = \begin{pmatrix} k & 0 \\ 0 & k^{-1} \end{pmatrix}$$

where $k \in \mathbb{Z}$ with $k \neq \pm 1$ (so λ_k is not unipotent) and $\gcd(k, M_p) = 1$ (so λ_k is invertible mod M_p).

Note: w is the standard Weyl element of order 4 in $\mathrm{SL}_2(\mathbb{Z})$; it replaces μ and satisfies $\det w = 1$. In the classical presentation of the modular group, set

$$T := \sigma = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad S := w^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix};$$

then $\mathrm{SL}_2(\mathbb{Z}) = \langle S, T \mid S^4 = I,\ (ST)^3 = S^2 \rangle$ (so $\langle \sigma, w \rangle = \mathrm{SL}_2(\mathbb{Z})$). This identification makes the Zariski-density of $\langle \sigma, w \rangle$ immediate: $\mathrm{SL}_2(\mathbb{Z})$ is a lattice in $\mathrm{SL}_2(\mathbb{R})$ and hence Zariski-dense. Adding λ_k with $k \neq \pm 1$ preserves Zariski-density (the group only grows) while ensuring the generator set is not contained in a unipotent or torus subgroup.

Lemma 4.15 (Zariski density of $\langle \sigma, w \rangle$)**.** *The group $G_0 = \langle \sigma, w \rangle = \mathrm{SL}_2(\mathbb{Z})$ is Zariski-dense in SL_2.*

Proof. $\mathrm{SL}_2(\mathbb{Z})$ is a lattice in $\mathrm{SL}_2(\mathbb{R})$, hence Zariski-dense in $\mathrm{SL}_2(\mathbb{C})$. Setting $T = \sigma$ and $S = w^{-1}$, the standard presentation $\langle S, T \mid S^4 = I,\ (ST)^3 = S^2\rangle$ is verified in §4.7. □

Remark. *Adding* $\lambda_k = \left[\begin{smallmatrix} k & 0 \\ 0 & k^{-1} \end{smallmatrix}\right]$ *for* $k \in \mathbb{Z} \setminus \{0, \pm 1\}$ *with* $\gcd(k, 6) = 1$ *preserves Zariski-density (the group only grows). It additionally ensures the generator set is not confined to a unipotent or torus subgroup, which is relevant when* λ_k *is used in the prime-lifted tonal action (Definition 4.14).*

4.8 Reflective versus Rotational Tonal Inversion

We distinguish two inversion types that arise in tonal systems. Call $r : p \mapsto -p$ (pitch-class reversal on $\mathbb{Z}/n\mathbb{Z}$) a *reflective* inversion, and call the Weyl rotation $(b, e) \mapsto (e, -b)$ on $\mathbb{Z}^2$ a *rotational* inversion. Both are called "inversion" in common usage, but they act on spaces of different dimension and land on opposite sides of Theorem 4.10.

Definition 4.16 (Reflective inversion)**.** *A* reflective inversion *on a cyclic pitch space* $\mathbb{Z}/n\mathbb{Z}$ *is an automorphism* r *satisfying*

$$r^2 = \mathrm{id}, \qquad r\sigma r^{-1} = \sigma^{-1}.$$

The standard example is $r(p) = -p \pmod{n}$ *(pitch-class inversion). The symmetry group* $\langle \sigma, r\rangle$ *is the dihedral group* D_n*.*

Definition 4.17 (Rotational inversion)**.** *A* rotational inversion *compatible with* σ *is an automorphism* $w \in \mathrm{SL}_2(\mathbb{Z})$ *with* $w^2 = -\mathrm{id}$ *and* $\det(w) = 1$*. Setting* $S := w^{-1}$ *and* $T := \sigma$*, we require the modular relation*

$$S^4 = I, \qquad (ST)^3 = S^2.$$

These are the standard $\mathrm{SL}_2(\mathbb{Z})$ *presentation relations (Lemma 4.18); they tie* w *to the translation direction* σ*.*

Remark. *For the canonical* $\sigma = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ *and* $w = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$*,*

$$w\sigma w^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} =: \sigma_{\downarrow},$$

so w *conjugates* σ *(upper unipotent) to* $\sigma_{\downarrow}$ *(lower unipotent): a harmonic role exchange, not inversion to* σ^{-1}*. The modular relation* $(ST)^3 = S^2$ *is the correct algebraic tie. The Weyl element* $w = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$ *(acting as* $(b, e) \mapsto (e, -b)$*) is the canonical example.*

Lemma 4.18 (Harmonic lattice generation)**.** *Let* $\sigma = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ *and* $w = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$*. Then* $w^2 = -\mathrm{id}$*,* $\det(w) = 1$*, and* $\langle \sigma, w \rangle = \mathrm{SL}_2(\mathbb{Z})$*.*

Proof. Set $S := w^{-1} = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ and $T := \sigma$. Direct matrix computation gives $S^4 = I$ and $(ST)^3 = S^2 = -I$ (see §4.7). The presentation $\langle S, T \mid S^4 = I,\ (ST)^3 = S^2 \rangle$ is a classical generator theorem for $\mathrm{SL}_2(\mathbb{Z})$ [2]. Hence $\langle \sigma, w \rangle = \langle T, S \rangle = \mathrm{SL}_2(\mathbb{Z})$. □

Proposition 4.19 (Tonal Inversion Dichotomy)**.** *Let* $\mathcal{T}$ *be a tonal system closed under semitone transposition* σ*.*

(A) *If* $\mathcal{T}$ *admits a* reflective *inversion* r *on a* 1*-dimensional pitch space* $\mathbb{Z}/n\mathbb{Z}$*, then* $G_{\mathcal{T}} = \langle \sigma, r \rangle \cong D_n$ *is virtually abelian: polynomial ball growth, Case (A) of Theorem 4.10.*

(B) *If* $\mathcal{T}$ *admits a* σ*-compatible rotational inversion* w *(Definition 4.17), then by Lemma 4.18,* $\langle \sigma, w \rangle \cong \mathrm{SL}_2(\mathbb{Z})$*; in particular* $G_{\mathcal{T}}$ *is Zariski-dense: exponential ball growth, Case (B) of Theorem 4.10.*

The regime is determined by the inversion type: reflective ($r^2 = \mathrm{id}$*,* 1*-dimensional) yields Case (A);* σ*-compatible rotational (*$S^4 = I$*,* $(ST)^3 = S^2$*) yields Case (B).*

Proof. Case (A). D_n is virtually abelian; Theorem 4.10(A) applies.

Case (B). By Lemma 4.18, $\langle \sigma, w \rangle = \mathrm{SL}_2(\mathbb{Z})$. Apply Theorem 4.10(B) and Corollary 4.11. □

Remark (Dimensional elevation)**.** *Reflective inversion is an order-*2 *isometry of the pitch circle* S^1*; it preserves the one-dimensional geometry. Rotational inversion cannot be realised as an isometry of* S^1*: it requires a* 2*-dimensional harmonic lattice. The passage from reflective to rotational inversion is therefore a dimensional elevation from* S^1 *to the hyperbolic plane* $\mathbb{H}$ *(on which* $\mathrm{SL}_2(\mathbb{Z})$ *acts via Möbius transformations), and this elevation is precisely what forces the exponential growth regime.*

4.9 Worked Examples — Corrected

Example A — Virtually Abelian (Case A). Let $\Gamma = \{\pm\sigma\}$ where $\sigma = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$. Then $G = \langle \sigma \rangle \cong \mathbb{Z}$ is abelian, hence virtually solvable.

The Cayley graph $\mathcal{G}_p = \mathrm{Cay}(\mathbb{Z}/p\mathbb{Z}, \{\pm 1\})$ is a cycle of length p. Its spectral gap $\gamma_p = 1 - \cos(2\pi/p) \sim 2\pi^2/p^2 \to 0$. Its diameter is $\lfloor p/2 \rfloor \sim p/2$.

This family is *not* an expander: $\gamma_p \to 0$. Rescaling by $s_p = p$ gives convergence to the circle S^1: a smooth continuous limit exists. *Continuous-time behavior is possible.*

Example B (Case B: Zariski-dense). Let $\Gamma = \{\sigma^{\pm 1}, w^{\pm 1}\}$ as in Definition 4.14. Then $G = \mathrm{SL}_2(\mathbb{Z})$ (Zariski-dense lattice).

By Bourgain–Gamburd [1] (Theorem 4.7(B)(iii)), the Cayley graphs $\mathcal{G}_p = \mathrm{Cay}(\mathrm{SL}_2(\mathbb{F}_p), \rho_p(\Gamma))$ form an expander family with uniform gap $\gamma_p \geq \varepsilon > 0$.

Diameter: $\mathrm{diam}(\mathcal{G}_p) = O(\log p)$. Ball volume: exponential in graph distance ($|B(x,r)| \geq e^{cr}$, Theorem 4.10(B)). No rescaling produces any polynomial-volume limit (Corollary 4.11). *Harmonic motion is geometrically rigid.*

4.10 Structural Summary

$$\begin{array}{lclcl} G \text{ virt. solvable} & \Longleftrightarrow & \text{poly. ball growth} & \Rightarrow & \text{poly-vol limit exists} \\ G \text{ Zariski-dense} & \Longleftrightarrow & \text{exp. ball growth} & \Rightarrow & \text{no poly-vol limit} \end{array}$$

The growth class (polynomial vs. exponential) is the structural invariant, determined entirely by the Zariski closure of $G \leq \mathrm{SL}_2(\mathbb{Z})$. Spectral gap ($\gamma_p$) is a consequence of Zariski-density, not the primary object; it is used as an intermediate tool (Bourgain–Gamburd $\Rightarrow$ Cheeger $\Rightarrow$ exponential growth) rather than the dichotomy's engine.

4.11 Tonal Consequence

The following corollary instantiates the abstract dichotomy in concrete musical terms, using the generators identified in Definition 4.14.

Corollary 4.20 (Tonal Regime Dichotomy)**.** *Under the prime-lifted tonal action (Definition 4.14):*

(A) ***Transposition-only regime.*** *Let $\Gamma = \{\sigma^{\pm 1}\}$ (pure semitone transposition), so $G \cong \mathbb{Z}$. Restricting to any $\langle\sigma\rangle$-orbit, the Cayley graph is a cycle C_{M_p} of length $M_p = 24p$. Hence $\gamma_p \to 0$ and the rescaled cycle converges to S^1 (cf. Example A). Harmonic motion in this regime is* geometrically smooth*: it embeds in a continuous Riemannian flow.*

(B) ***Harmonic inversion regime.*** *Let $\Gamma \supseteq \{\sigma^{\pm 1}, w^{\pm 1}\}$ with w a rotational inversion (Definition 4.17). Proposition 4.19(B) gives $G = \mathrm{SL}_2(\mathbb{Z})$. Ball growth in $\{\mathcal{G}_p\}$ is exponential; by Corollary 4.11, no polynomial-volume scaling limit exists. Harmonic motion is* geometrically rigid*: no rescaling of reachability time produces a Riemannian polynomial-volume scaling limit.*

Proof. Case (A): $G \cong \mathbb{Z}$, virtually abelian; Theorem 4.10(A) and Example A. Case (B): $G = \mathrm{SL}_2(\mathbb{Z})$ by Proposition 4.19(B); Theorem 4.10(B) and Corollary 4.11. □

Remark (Musical interpretation)**.** *The dichotomy is a property of the* generator regime*, not of individual chord moves. A harmonic system whose generator regime is transposition-only has abelian lift ($G \cong \mathbb{Z}$); its large-scale geometry is a circle, admitting smooth voice-leading flows. A system that combines transposition with inversion generates* $\mathrm{SL}_2(\mathbb{Z})$*, whose congruence Cayley graphs have* exponential ball growth*. The Generator Growth Dichotomy says:* exponential growth is incompatible with any polynomial-volume scaling limit. *No rescaling of reachability time can produce a smooth, Riemannian, or sub-Riemannian description of the large-scale harmonic motion.*

References

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[3] C. R. Matthews, L. N. Vaserstein, and B. Weisfeiler. Congruence properties of Zariski-dense subgroups of semisimple algebraic groups. *Proceedings of the London Mathematical Society*, 48(3):514–532, 1984.

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